INCH-POUND

MIL-P-53051A(ME)
23 September 1994
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25 February 1985 and
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11 June 1987

MILITARY SPECIFICATION

PUMPING ASSEMBLY, WHEEL-MOUNTED, DIESEL-ENGINE-DRIVEN, SELF-PRIMING, CENTRIFUGAL, 600 GPM AT 350 FEET TOTAL HEAD AND 1250 GPM AT 180 FEET TOTAL HEAD

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a wheel-mounted, diesel-engine-driven, self-priming centrifugal, 6-inch hoseline pump and a wheel-mounted, diesel-engine-driven, self-priming centrifugal, flood and transfer pump.

1.2 Classification. The pumping assembly shall be one of the following classes as specified (see 6.2).

Class I	-	Six-inch hoseline; 600 gallons per minute (gpm) at 350 feet total head when pumping water, wheel mounted, pressure regulated, for drinking water.
Class II		Flood and transfer; 600 gpm at 350 feet total head and 1,250 gpm at 180 feet total head when pumping 1.0 specific gravity liquids, wheel-mounted, for fuel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD, STE 104, FT BELVOIR, VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4320

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and, standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-C-390	-	Copper Alloy Castings.
VV-F-800	-	Fuel Oil, Diesel.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.

MILITARY

MIL-V-173	-	Varnish, Moisture and Fungus Resistant (For Treatment of Communications Electronics and Associated Equipment).
MIL-P-514	-	Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-L-2104	-	Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-G-3859	-	Grease Gun, Hand Operated, Lever, Push and Screw Type.
MIL-T-5624	-	Turbine Fuel, Aviation, Grades JP-4, JP-5 and JP-5/JP-8 ST.
MIL-H-83794	-	Hose, Rubber, Hydraulic, Fuel and Oil Resistant.
MIL-E-10062	-	Engine, Preparation for Shipment and Storage of.
MIL-C-10387	-	Coupling, Clamp, Pipe with Bolts and Synthetic Rubber Gasket for Grooved Pipe and Tube.
MIL-P-106O3	-	Pumps and Pumping Units, Centrifugal, Power Driven, for Water, Packaging of.
MIL-G-10924	-	Grease, Automotive and Artillery.
MIL-P-14105	-	Paint, Heat-Resisting (for Steel Surfaces).
MIL-P-15024	-	Plate, Tags and Bands for Identification of Equipment.
MIL-P-15024/9	-	Plate, Tags and Bands, Aircraft Loading Data Plate.
MIL-G-23827	-	Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-A-46153	-	Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-L-46167	-	Lubricating Oil, Internal Combustion Engine, Arctic.

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MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176 - Brake Fluid, Silicon, Automotive, All Weather, Operational and Preservative.
MIL-E-52649 - Engine, Cold Starting Aids, Ether Fuel Primers.
MIL-C-53039 - Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-C-62122 - Cable Assembly, Slave, Electrical Connectors, Plug and Receptacle.
MIL-V-81940 - Valve, Sampling and Bleed, Hydraulic, Type II Systems.
MIL-V-81940/2 - Valve, Sampling and Bleed, Hydraulic, Type II Systems NPTF Fitting.
MIL-T-83133 - Turbine Fuels, Aviation, Kerosene Types, NATO F-34 (JP-8) and NATO F-35.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.
FED-STD-595 - Colors Used in Government Procurement.

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1366 - Packaging, Handling, Storage, and Transportation System Dimensional Constraints, Definition of.
MIL-STD-1410 - Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474 - Noise Limits for Military Materiels.
MIL-STD-1791 - Designing for Internal Aerial Delivery in Fixed Wing Aircraft.
MS24532 - Voltmeter, Battery-Generator, 28 Volt DC.
MS24543 - Indicator, Temperature, Electrical Resistance: 24 Volt DC, Range 120 - 240 °F.
MS27022 - Coupling Half, Quick Disconnect, Cam-locking Type, Male, External Pipe Thread, Type III.
MS27028 - Coupling Half, Quick Disconnect, Cam-locking Type, Cap, Dust, Type IX.
MS35643 - Gasket: Synthetic Rubber, Cap, Fuel Tank.
MS35644 - Filler Neck: Fuel Tank-Military Vehicles.

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MS35645 - Cap, Fuel Tank: Tactical Vehicle, Fording.
MS52125 - Composite Light-Tail, Stop, Turn and Marker.
MS52149 - Batteries, Storage, Lead Acid, (Low Maintenance).
MS75020 - Connector, Plug , Electrical-12 Contact, Intervehicular, 24 Volt Waterproof.

(Unless otherwise indicated, copies of federal and militaary specifications, standards, and handbooks are available from: STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

13228E9844 - Information Plate, 600 GPM.
13230E5670 – Gauge, Compound.
13230E5671 - Gauge, Pressure.

Del) PROJECT MANAGER - MOBILE ELECTRIC POWER

69-657 - Paddle Lock.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from: US ARMY BELVOIR RDE CTR, ATTN SATBE FSH, 10101 GRIDLEY RD, STE 104, FT BELVOIR VA 22060-5818.)

PUBLICATIONS

FOOD AND DRUG ADMINISTRATION (FDA)

Code of Federal Regulations, Title 21, Chapter 1, Part 177.

(The Code of Federal Regulations [CFR] and the Federal Register [FR] are for sale on a subscription basis by: SUPT OF DCMNTS, GVT PRINTG OFC, WASHINGTON DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 48	-	Gray Iron Castings.
B 26	-	Aluminum Alloy Sand Castings.
B 584	-	Copper Alloy Sand Castings for General Applications.
D 3951	-	Standard Practice for Commercial Packaging.

(Application for copies should be addressed to: AMERCN SCTY FOR TEST & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to: AMERCN SCTY OF MECHL ENGRS, 345 E 47TH STRET, NEW YORK NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

D1.1		Structural Welding Code - Steel.

(Application for copies should be addressed to: AMERCN WELD SCTY, 550 NW LEJEUNE RD, PO BOX 351040, MIAMI FL 33135.)

HYDRAULIC INSTITUTE

Hydraulic Institute Standards for Centrifugal, Rotary and Reciprocating Pumps.

(Application for copies should be addressed to: HYDRLC INST, 14600 DETROIT AVE, 712 LAKEWOOD CTR N, CLEVELAND OH 44107.)

NATIONAL SANITATION FOUNDATION (NSF)

Standard 61	-	Drinking Water System Components - Health Effects.

(Application for copies should be addressed to: NATL SANITIN FNDTN, 3475 PLYMOUTH RD, PO BOX 1468, ANN ARBOR MI 48106.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J614	Marking For Oil Dipsticks.
J847	Full Trailer Tow Bar Eye.

(Application for copies should be addressed to: SCTY OF AUTOMTV ENGRS, INC, 400 CMNWLTH DR, WARRENDALE PA 15096.)

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SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to: S MGMT CNTRL DIST, 9150 E FLAIR DR, EL MONTE CA 91731.)

TIRE AND RIM ASSOCIATION, INC.

Yearbook.
Supplementary Service Data Book.

(Application for copies should be addressed to: TIRE & RIM ASSN, INC, 3200 W MARKET STRET, STE 304, AKRON OH 44313.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The class I pumping assembly shall be a wheel-mounted, diesel-engine-driven, self-priming centrifugal pumping assembly. The pumping assembly shall have an operational point of 600 gpm at 350 feet total head when pumping drinking water. The pumping assembly shall be equipped with automatic pump/engine controls. The class II pumping assembly shall be a wheel-mounted, diesel-engine-driven, self-priming centrifugal pumping assembly. The pumping assembly shall have an operational point of 1,250 gpm at 180 feet total head when pumping liquids with a 1.0 specific gravity.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.3 Materials. The material shall be as specified herein. When a definite material is not specified, a material shall be used which will enable the pumping unit to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product. The material shall have no adverse effect on the health of personnel when used for its intended purpose. The finished class I pumping assembly shall contain no materials or substances which might leak out or disintegrate and cause the pumped water to become nonpotable. The lines and all surfaces that contact

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potable water shall conform to the Federal Food, Drug and Cosmetic, Food Additives, Chapter I, Title 21, Part 177, Code of Federal Regulations for contact with food or shall be tested and approved for use with potable water in accordance with NSF Standard 61. The compounds listed in section 177.1020, 177.1030, 177.1040, 177.1050, and 177.1480 of the Federal Food, Drug and Cosmetic, Indirect Food Additives, Chapter I, Title 21, Code of Federal Regulations shall not be acceptable materials. Questions pertinent to this effort shall be referred by the procuring activity to the departmental medical service who will act as advisor to the procuring agency.

3.3.1 Material deterioration Prevention and control. The pumping assembly shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments (see 3.7.1 and 3.7.2) to which the pumping assembly may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the pumping assembly may be newly fabricated from recovered materials to the maximum extent practicable, provided the pumping assembly produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the pumping assembly.

3.3.4 Cadmium Plating. Except for grade 5B and grade 8 fasteners, cadmium plating shall not be used.

3.4 Manufacturer’s Product lines. The pump and engine and all their components shall be the latest models of the contractor (and any subcontractors). The pumping assembly shall be equipped with all components necessary to meet the requirements stated herein.

3.5 Design. The pumping assembly shall be designed to permit easy accessibility for maintenance and service in the field (see 3.9). The design shall prevent conditions hazardous to personnel or deleterious to equipment.

3.5.1 Ease of maintenance. All major assemblies and installed attachments shall be accessible for repair, replacement, and adjustment without the removal of other major assemblies and installed attachments not usually removed. Covers, plates, and mounting brackets which must be removed for component adjustment, repair, or replacement shall be equipped with quick-disconnect

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fasteners (no tools required, maximum one complete turn to release for rotary operated). Only flat or pan head fasteners shall be used. Various size fasteners shall be readily distinguishable. Drain outlets shall be located for accessibility. All drain plugs and filler caps shall be durable metal and shall be attached to the unit with permanent retaining chains or cables which shall not impede removal or replacement of the plugs or caps from their receptacle. A maintenance schedule shall be furnished prior to the start of any testing and an operator's field maintenance chart shall be affixed to the pumping assembly per 3.9. Design for maintainability shall be as specified in 3.9.

3.5.2 Reliability. The specified mean-time-between-failures shall be at least 200 hours when the pumping assembly is tested in accordance with 4.6.2.13.

3.6 Threads. Screw threads shall be in accordance with FED-STD-H28 .

3.7 Environmental.

3.7.1 Operating temperature. The class I pumping assembly shall perform as specified herein in any ambient temperature of +135 to +32 CF. The class II pumping assembly shall perform as specified herein in any ambient temperature from +135 to -25 °F.

3.7.2 Storage temperature. The class I and class II pumping assemblies shall not be damaged by storage in any ambient temperature from +160 to -65 °F.

3.8 Safety. All shafts or rotating parts of the pumping assembly shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust components exposed to contact by personnel or which create a fire hazard, shall be fully guarded or insulated. The fuel fill port shall be located to allow refueling without first opening any panels or doors. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Exhausts or discharges from the equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded and grounded to prevent short circuit or sparking.

3.9 Human factors engineering. The pumping assembly shall conform, but not be limited to, the following human factors engineering design criteria as specified in MIL-STD-1472: General requirements (4); control/display integration (5.1); visual displays (5.2); controls (5.4); labeling (5.5); anthropometry (5.6); design for maintainability (5.9); and hazards and safety (5.13).

3.10 Noise limits. Steady-state noise produced by the pumping assembly shall conform to MIL-STD-1474, Category "D" equipment requirements, with the exception of MIL-STD-1474, 5.2, 5.3, and 5.4,

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when tested in accordance with MIL-STD-1474 and 4.6.2.8. Noise levels shall not exceed 85 dB(A) at the measurement locations defined as follows: Per MIL-STD-1474, 5.1.2.1.2, occasionally occupied positions shall be 39 inches horizontally from the basic envelope of the unit and a vertical distance of 65 inches above ground level. The operator's position shall be 18 inches horizontally from the center of the control/instrument panel.

3.11 Pump.

3.11.1 Performance. The pump shall be a 6-inch integrally self-priming (after case has been initially filled) centrifugal pump. The class I pump shall have a rated capacity of 600 gpm at 350 feet total head when pumping water. The class II pump shall meet the class I performance requirements and also have a rated capacity of 1,250 gpm at 180 feet total head when pumping fresh water.

3.11.1.1 Tilted position. The pumping assembly shall deliver its rated capacity when positioned at any angle up to 15 degrees from the horizontal in any direction.

3.11.1.2 Priming. The pump shall prime and produce rated capacity within 2 minutes on a 10-foot static suction lift after initial filling of the pump casing.

3.11.2 Construction.

3.11.2.1 Class I pump case and volute. The class I pump case and volute shall be a one-piece gray iron casting conforming to ASTM A 48, class 30. The cut water shall be a gray alloy casting conforming to ASTM A 48, class 30S, and shall be integrally cast into the pump case. The case shall have at least two mounting feet, a valve for complete pump drainage in 1 minute, and a 1-1/2 inch priming port with a check valve, coupling, and dust cap.

3.11.2.2 Class II pump case and volute. The class II pump case and volute shall be a one-piece aluminum casting conforming to ASTM B 26, alloy 356.0-T6. The cutwater shall be a gray iron alloy casting conforming to ASTM A 48, class 30S, and shall be integrally cast into the pump case. The case shall have at least two mounting feet, provisions for complete pump drainage, and a 1-1/2 inch priming port with a connector check valve to prevent pumpage from flowing out of port, coupling, and dust cap.

3.11.2.3 Impeller. The impeller shall be fabricated of bronze conforming to ASTM B 584, copper alloy UNS no. C83600. The wear ring shall be copper alloy no. 836 conforming to QQ-C-390.

3.11.2.4 Seals. The pump shall be equipped with self-lubricating mechanical seals that will not be damaged or compromised by operation of the pumping assembly in any operating condition specified herein.

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3.11.2.5 Priming port. The pump shall be equipped with a 1-1/2 inch priming port fitted with a quick-disconnect, cam-locking type male coupling conforming to MS27022-9. The coupling shall have a cam-locking type dust cap conforming to MS27028-9 with a short length of chain securing it to the pump. A check valve shall be installed between the priming port coupling and the pump housing to prevent back flow from the pump casing.

3.11.2.6 Piping and fittings. All piping, valves, and fittings on the suction and discharge of the class I pump shall be flanged and fabricated from ASTM A 48, class 30, gray iron, except as specified herein. All piping, valves, and fittings on the suction and discharge of the class II pump shall be flanged and fabricated from 356.0-T6 aluminum, except as specified herein. All piping and fittings shall be mounted securely and shall withstand an internal working pressure of 1.5 times the shutoff pressure at rated speed without leaking or becoming deformed. The discharge piping shall contain a 1/4-inch (NPT) vent valve to remove air during priming. The vent shall be located at the highest point in the discharge piping and the valve shall be routed to the exterior of the pump and engine housing for easy accessibility.

3.11.3 Class I pump accessories. The class I pump shall be furnished with 6-inch single grooved connections, suction and discharge, that accept a coupling conforming to MIL-C-10387. A coupling conforming to MIL-C-10387 and a blind adapter shall be furnished for both the suction and discharge connection to act as cover to prevent entry of foreign material. The coupling and gasket shall conform to the requirements herein for lines and surfaces that contact potable water (see 3.3).

3.11.4 class II pump accessories. The following accessories shall be furnished with the class II pump:

a. Six-inch single grooved flanged adapters, suction and discharge, that accept a coupling conforming to MIL-C-10387. The flanged adapters shall be aluminum castings conforming to ASTM B 26, alloy 356.0-T6.
b. Discharge check valve, having a 6-inch flanged housing. The check valve shall be an aluminum casting conforming to ASTM B 26, alloy 356.0-T6.
c. Six-inch gate valves, discharge and suction. The valves shall be positioned to accept both hose and pipe. The discharge and suction valves shall be supported by a bracket connected to the pumping unit frame.
d. Suction strainer assembly, 1/4-inch mesh in–line type, reinforced basket strainer constructed of type 304 stainless steel . The basket shall be removed from the strainer assembly without disassembly or removal of other components and without the benefit of tools. The strainer body and lid shall be an aluminum casting conforming to ASTM B 26, alloy 356.0-T6.

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e. Air eliminator - An automatic air eliminator assembly shall be located at the highest point on the discharge side of the pump to eliminate air from the pump during priming.

3.12 Engine. The engine shall be an industrial, diesel engine selected in accordance with MIL-STD-1410, class II. The engine shall deliver power at its continuous duty rating to meet the pump performance requirements of 3.11.1. The engine shall be capable of operating on fuels conforming to VV-F-800, MIL-T-5624 and MIL-T-83133 . The engine shall be equipped with but not limited to the following:

a. A 24-volt electric cranking system with a charging alternator and batteries suitable for -25 l F starting.
b. Extra heavy duty extended life air cleaner, dry type, replaceable cartridge, with restriction indicator.
c. Fuel filters, primary, water separating type; secondary, each with a drainage provision which is located such that filters can be drained to a container.
d. Manual engine controls to include a waterproof pushbutton starter switch and engine run/stop switch.
e. Automatic engine shutdown controls in the event of engine overspeed, high coolant temperature, or low lubricant oil pressure. All controls shall have manual resets.
f. Exhaust system with guard and a pressure actuated rain cap which extends through the top of the pump and engine housing.
g" Drain tube to permit draining the engine oil into a container without soiling other components of the system. Drainage shall be easily accessible and through the outer housing.
h. Auxiliary charging receptacle conforming to MIL-C-62122 for recharging the batteries from a remote power source without operation of the engine. The auxiliary charging receptacle shall come through the housing and be labeled according to 13228E9844-23.
i. Cold weather starting aid conforming to MIL-E-52649, type 3.
j. Valve to sample engine oil for testing: Sample must be taken from pressure side of engine. Sampling from sump is not acceptable. The valve shall conform to MIL-V-81940 and MIL-V-81940/2, part no. M81940/2-1.
k. Instrumentation consisting of lubricating oil pressure gauge, coolant temperature gauge, tachometer-hourmeter, battery charging indicator, an air cleaner restriction indicator and fuel gauge.
l. Heavy duty cooling system capable of maintaining the manufacturer’s recommended operating temperature when the pumping assembly is performing at rated capacity in the environmental conditions (see 3.7). If water cooled, the radiator shall have provisions to be cleaned from the front using compressed air.
m. Engine fan belt guard that will provide quick access to and replacement of belt without removal of guard or mounting screws.

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n. Oil dip stick conforming to SAE J614.
o. A muffler which has provisions for draining combustion produced water to the exterior of the pumping assembly below the bottom surface of the pump and engine housing.

3.13 class I pump automatic pump controller. The class I pumping assembly shall be equipped with an automatic pump controller which will reduce engine speed to limit high discharge pressure or low suction pressure. The controller shall sense the discharge and begin to act at 150 pounds per square inch (psi) by reducing engine speed. If the pressure continues to increase, the engine speed will be continuously reduced until idle setting is reached. The controller shall also sense the pressure on the suction side of the pump. If the inlet pressure falls below 20 psi the controller shall start to reduce engine speed. If the inlet pressure continues to fall when the pressure reaches 10 psi, the engine speed will be reduced to idle. The controller shall be capable of being turned off so that the engine speed will be regulated by the engine governor via the hand throttle.

3.13.1 Operator controls. The operator controls portion of the automatic pump controller shall have two controls. One control shall be a three-position rotary switch providing selection of the desired mode of operation; manual, automatic pressure control or automatic speed control. The second operator control knob shall be a combination switch and rheostat. From fully counterclockwise to fully clockwise the switch/rheostat position and ranges shall be: (1) start range, and (2) speed set/discharge pressure set range. Fully clockwise, the switch/rheostat knob shall provide maximum engine speed or maximum discharge pressure depending on the position of the three-position rotary switch. The two operator controls shall be mounted on the instrument panel and identified as specified by figure lb. These two operator controls shall not be included on the class II pump control and instrument panel.

3.13.2 Hardware. The pressure sensing and speed command components shall be all solid state electronics. They shall operate an electromechanical actuator to precisely position the fuel control , on the engine, for isochronous (zero droop) speed setting. Military type quick disconnect electrical connectors (see 3.9) shall be used for the cables that connect to the automatic pump control modules.

3.13.3 Pressure lines. The pressure lines that provide suction and discharge pressure to the respective transducers in the automatic pump control shall have drainage cocks. The drainage cocks shall be located to allow complete drainage of the pressure transducers to prevent damage during cold storage. The suction line and associated transducer shall be capable of withstanding at least 150 psig hydrostatic pressure. The discharge line and associated transducer shall be capable of withstanding at least 1.5 times the shutoff pressure at rated speed. The transducers shall be located to allow for complete drainage to prevent damage during cold storage.

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3.14 Control and instrument panel. The aluminum instrument panel shall be covered by a door which is hinged on the left side, has automatically engaging hold open devices and latches closed. All operator controls shall be on the panel covered by the door. The instrument panel (which is located behind the door) shall be hinged on the right hand side and on the left hand side shall be secured by 1/4 turn screwdriver slot type fasteners. The automatic pump control electronics of the class I pump shall be contained in the space behind, but not mounted on, the instrument panel. The class II pump control and instrument panel shall not include the two automatic pump controls. All components shall be marked on the rear of panel with identification such as "Sw#2" for switch #2, "G#2" for gauge #2, "P#2" for plug #2, "J#2" for jack #2, etc. The instrument panel shall conform to figures la, lb and 1c, and include as a minimum the following:

a. Compound gauge conforming to drawing 13230E5670.
b. Pressure gauge conforming to drawing no. 13230E5671.
c. Tachometer-hourmeter, 0 to 3000 rpm, either mechanical or electrical.
d. Adjustment screws shall have wrenching recesses or slot filled with a plastic material to prevent indiscriminate adjustment,
e. Voltmeter conforming to MS24532.
f. Engine oil pressure gauge.
g. Engine low oil pressure shutdown with sunlight compatible LED indicator light and a manual reset.
h. Engine overspeed shutdown with sunlight compatible LED indicator light and a manual reset.
i. Engine overheated shutdown with sunlight compatible LED indicator light and a manual reset.
j. A three-position switch indicating stop, run and start. The low oil pressure shutdown shall be overcome by the start position of the switch to permit starting. When released from the start position, the switch shall automatically return to the run position.
k. Fuel gauge graduated in 1/4 increments from empty to full.
l. Panel shall be illuminated in accordance with 3.9. Lights shall include a dimmer switch which allows adjustment from completely off to full illumination. Lights shall not extend more than one inch horizontally from the vertical plane of the instrument panel. There shall be a switch which automatically cuts off the panel lights when the control and instrument panel door is closed.
m. Air filter restriction indicator.
n. Control for the ether injection system.
o. Manual throttle.
p. Engine temperature gauge conforming to MS24543.

3.14 .1 Wiring. All wiring connections shall use ring lugs on screw posts with nuts, and lockwashers. All electrical terminals outside the instrument panel shall have standoff non-conductive covers to shield against damage, inadvertent shorting, and weather. All battery and starting system connections shall be equipped with

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rubber boots. The positive battery terminal cable shall be red and labeled with a "+", and the negative battery cable shall be black and labeled with a "-". All wiring shall be gathered, routed, and securely clamped to prevent excessive movement and inadvertent damage during maintenance procedures. Grounding of all electrical components shall be done at a single point. Military type quick disconnect electrical connectors, per 3.9, shall be used, as applicable. All electrical components shall conform to MIL-STD-454.

3.15 Fuel system. The fuel tank shall have sufficient capacity to provide 8 hours operation at rated performance without refueling. Two 3-way fuel selector valves, two 10-foot long fuel hoses conforming to MIL-H-8794, and storage on the pumping assembly for the hoses shall be furnished to permit engine operation from the fuel tank or an auxiliary fuel service. The valves shall be labeled in accordance with 13228E9844-16 and 13228E9844-17. The fuel tank fill shall be routed to an external fuel fill port. The fuel fill port shall be located in a 45 degree recessed area on the exterior of the pump and engine housing and be designed such that pouring from a fuel can does not result in spillage. The fuel fill port shall be equipped with a filler neck conforming to MS35644, a tank cap conforming to MS35645, and a gasket conforming to MS35643. A 3-inch clearance shall be provided between the tank cap and the sides of the recessed area. An information plate conforming to drawing 13228E9844-19 shall be attached at the base of the recessed surface just below the fuel tank cap. The bottom surface of the fuel tank shall be protected by a metal guard. No fuel spillage shall occur when the fuel tank is full and the pumping assembly is operating and tilted as described in 3.11.1.1.

3.16 Wheeled mounting. The chassis shall have two single wheels on an axle assembly attached through leaf springs to the underside of the pump frame. The axle and chassis assembly shall have a rated capacity exceeding the maximum payload, but not less than 6,000 pounds and be equipped with a minimum of two shock absorbers for stability.

3.16.1 Tires and wheels. Tires and wheels shall be in accordance with the guidelines of the Tire and Rim Association and shall be of sufficient size and durability to support loads in off-road conditions with no more than 50 psi air pressure. The tires shall contain tubes. Tire pressure shall be indicated directly above the tires "T.P. 50 PSI" in 1-inch high letters. Tires and tubes furnished shall be not more than 12 months old on the date of pumping assembly shipment.

3.16.2 Brakes. Brakes shall be of the hydraulic inertia type. Each pumping assembly shall be equipped with a parking hand brake with snap-over-center locking design. The force required to apply the parking brake to keep the trailer from moving down a 25 percent slope shall not exceed 35 pounds. The parking brake handle shall be attached to the side of the tongue and mounted such that the parking brakes are applied when the handle is in the up position. The parking brake shall be mounted on the curb side of the pumping

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assembly such that there are no obstructions for the operator through the entire range of handle motion. Brake lines and cables shall be routed so as not to interfere with the free movement of other trailer assembly components. The brake lines and cables shall be securely clamped along the routing path and routed to avoid brush and debris damage.

3.16.3 Towing extension (tongue). The tongue shall have a nominal height of 26 inches from the ground when the unit is level and parked on a level surface. The tongue shall have a towing eye in accordance with SAE J847. Weight on the towing eye with the pumping assembly parked on level ground shall be not more than 15 percent and not less than 10 percent of the gross weight of the unit. The tongue shall be equipped with an adjustable, hand-crank style jack incorporated into a tripod design landing gear to permit operation of the tongue by one person. The landing gear shall be equipped with a sand shoe with a minimum area of 150 square inches.

3.16.4 Stabilizing legs. The pumping assembly shall have two heavy duty hand-crank style stabilizing legs, one on each rear corner, with a sand shoe with a minimum of 150-square inch area to permit operation without tipping over when not attached to a towing vehicle. When not in use, the stabilizing leg shall swivel inward to a horizontal position and lock. When in the stored position, the stabilizing leg shall not protrude beyond the widest part of the trailer assembly. The storage and operational position of the stabilizing leg shall maintain the same swivel point. A label conforming to 13228E9844-33 and showing proper use during operation and transport shall be placed over each stabilizing leg.

3.16.5 Lights. The pumping assembly shall have stoplights and taillights in accordance with MS52125.

3.16.5.1 Electrical connections. The pumping assembly shall have intervehicular connections for the 24-volt electrical system with connectors conforming to MS75020-1.

3.17 Pump and engine housing. The pump and engine housing shall be durable and provide easy access to all controls and required maintenance areas. There shall be no protrusions from the basic envelope of the housing. All external parts, connections or controls must be recessed. All access doors shall be equipped with automatic engaging hold open devices as illustrated in figure 7 and paddle-type latches conforming to drawing 69-657. The overall dimensions of the pumping assembly shall conform to figures 2 and 3. Stencil specifications shall be as shown on figure 4. The actual configuration of the pump and engine housing shall be in accordance with figures 2, 3, 4 and 5.

3.17.1 Housing insulation. If acoustic absorbing material is used within the housing, that material shall be non-combustible and shall not absorb fuel or oil. Additionally, all exposed surfaces of acoustic absorbing material shall be enclosed by perforated metal covering and that covering shall be painted in accordance with 3.23.

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3.18 Rock guard. The fuel tank shall be protected from stones when the pumping assembly is towed.

3.19 Transportability. The pumping unit shall be capable of being transported (see 6.2), by military or commercial trailers, trains, marine vessels, aircraft and must be capable of withstanding the impact forces encountered in shipment without damage or permanent deformation. The pumping unit shall be equipped with tiedown or slinging provisions.

3.19.1 Tiedown previsions. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3, type IV equipment, and to MIL-STD-1791 for equipment restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified in 4.6.2.10 without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Tiedown points shall be labelled "TIEDOWN HERE" and combination lift/tiedown points shall be labelled "LIFT/TIEDOWN HERE" in accordance with figures 2 and 5.

3.19.2 Slinging provisions. The slinging provisions shall conform to MIL-STD-209, class 1 or 3, type IV equipment. The provisions shall enable the complete pumping unit to be lifted in the normal operating position. The provisions shall be located so that not less than 1-inch clearance is maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.19.1. Lift points shall be labelled "LIFT HERE" in accordance with figures 2 and 5.

3.19.3 Air transportability. The pumping unit shall meet the requirements of MIL-STD-1791 for air transport. The pumping unit shall be transportable by the C-130, C-141, C-5 aircraft and lifted by helicopters.

3.19.4 Rail transportability. The pumping unit shall be rail transportable in CONUS and NATO countries without restrictions. When loaded on a 50-inch (127 cm) high rail car, the pumping unit shall have a dimensional profile within the AAR outline diagram and the Gabarit International de Chargement (GIC) or PPI, as specified in MIL-STD-1366. The pumping unit shall be capable of withstanding shock loads resulting from rail impact testing specified in 4.6.2.13 without failure, damage, or permanent deformation.

3.20 Grounding assembly for class II pumping assembly. Two ground rods of 3/4-inch nominal diameter and 60 inches long of copper clad, cold-rolled steel shall be provided on the class II pumping assembly. Two uninsulated, braided ground wires, 10 American Wire Gauge, 6 feet long, shall be provided. A 50-ampere, uninsulated, electrical bronze, plier type clamp shall be soldered

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to each end of the wire. A ground terminal shall be bonded to the frame. A bronze mallet with a minimum of a 3-pound head shall be furnished for driving the rods into the ground.

3.21 Identification, instruction and shipping plates.

3.21.1 Identification plate. Each pumping assembly shall be identified in accordance with 13228E9844-26, on plates made of material conforming to MIL-P-514, type III, composition C, grade B, class 1. The plates shall be attached to the pumping assembly by screws, rivets, or bolts in a place where the plate will be visible and legible but not interfere with the operation and maintenance of the pumping assembly.

3.21.2 Identification of pump inlet and outlet. The pump suction and discharge shall be identified by markings integrally cast in the pump.

3.21.2.1 Identification of direction of flow. The inlet and outlet of the pumpcasing shall be marked with arrows denoting the direction of flow through the pump. The arrows shall be painted on and shall be black.

3.21.3 Instruction plates. In addition to labels required, each pumping assembly shall be equipped with instruction plates, including diagrams, warnings, and cautions describing any special or important procedures to be followed in assembling, operating, or servicing the pumping assembly and shall be in accordance with 3.9. The pumping assembly shall have an instruction plate to inform the operator on the basic information needed to operate the pumping assembly in both the automatic and manual modes. The left side of the plate shall have instructions for the manual mode, the right side for the automatic mode. Plates shall be made of a material conforming to MIL-P-514, type III, composition C, grade B, class 1. The class II pumping assembly plates shall include the warning "NO SMOKING WITHIN 50 FEET (15.2M)". The warning should be visible from a distance greater than 50 feet and visible from both sides of the pumping assembly. Each plate shall be attached to the pumping assembly by screws, rivets, or bolts in a place where the plate will be visible and legible, but not interfere with the operation and maintenance of the pumping unit.

3.21.4 Performance curve Plate. The class I pumping assembly performance curve plate shall be in accordance with 13228E9844-34, and attached to the pumping assembly by screws, rivets, or bolts in a location where the plate will be visible from the operator’s panel. The plate material shall be as specified in 3.21.3. The class II pumping assembly performance curve plate shall be attached to the pumping assembly by screws, rivets, or bolts in a location where the plate will be visible and legible. The class II pumping assembly performance curve plate shall have the same dimensions and material as the class I plate, however, the legend shall include psi for liquids of 0.72 specific gravity, 0.85 specific gravity, and 1.0 specific gravity.

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3.21.5 Shipping data plate. Each pumping assembly shall be provided with an aircraft loading data plate conforming to figure 3 of MIL-P-15024/9, type H, style 111. The plate shall be attached to the pump and engine assembly as specified in MIL-P-15024 by screws, rivets, or bolts where the plate will be visible and legible.

3.22 Lubrication. The pumping unit shall operate as specified herein when serviced with the following military lubricants, grades as applicable to the operating conditions:

Lubricating oil:	MIL-L-2104 or MIL-L-46167
Grease :	MIL-G-10924 or MIL-G-23827
Brake Fluid:	MIL-B-46176

The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated before delivery and shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.22.1 Lubrication fittings. The lubrication fittings shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Access is permissible via panels and plates equipped with hand-operable, quick-disconnect fasteners.

3.22.2 Pressure-release device. A pressure release device shall be provided where use of pressure lubricating equipment will damage grease seals.

3.23 Cleaning, treatment, and painting. Cleaning, treatment, and painting of the entire pumping assembly shall be in accordance with MIL-T-704, type F or G and as stated herein.

a. All external surfaces of the pumping assembly (except those that reach a temperature of 400 °F), regardless of the material selected, shall have a finish coat of CARC paint in accordance with MIL-C-46168 or MIL-C-53039. All marking and caution lettering shall be flat black. Any external surface that reaches a temperature of 400 °F during operation shall be coated with paint conforming to MIL-P-14105. Unless otherwise specified (see 6.2), color shall be tan 686A, color chip No. 33446 in accordance with FED-STD-595.
b. All other surfaces of the pumping assembly, to include those within the noise enclosure and those behind insulation material, shall be cleaned, treated, and painted using manufacturer’s commercial practices. The engine pump, internal components, hardware, fasteners, and fittings may be finished with the manufacturer's standard commercial paint and color, plating or treatment.

3.23.1 Camouflage pattern data. When specified (see 6.2), the contractor shall provide camouflage pattern data.

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3.24 Battery box. The battery box shall be made of 304 stainless steel not thinner than 11 gauge and shall be designed to rigidly restrain two 12-volt batteries as specified by MS52149-1. The height of this box shall be not greater than 20 percent of battery height. Batteries shall be restrained by electrically insulated clamping brackets secured by "J" bolts which clamp the batteries in place with captively retained wing nuts. The box shall contain four 1/2-inch holes for drainage. The drain holes shall be positioned so that they drain directly to the ground. Electrolyte as required shall be provided. The electrolyte shall be packaged as required for aircraft transport. A pictorial instruction plate showing battery connections shall be provided as specified by 13228E9844-30. The instruction plate shall be mounted on the front of the battery box .

3.25 Electromametic interference. The pumping assembly shall be equipped for the suppression of electromagnetic interference in accordance with MIL-STD-461, UM03 requirements for class Cl, group II equipment.

3.26 Fungus- and moisture-resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.26.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

a. Components or circuit elements that are inherently fungus- and moisture-resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

The varnish shall be applied by spray, brush, or a combination of both. A dry-film thickness of not less than 1 mil shall be applied to component or element surfaces. Surfaces where varnish is applied shall be free from all foreign matter which would interfere with the adherence or function of the varnish.

3.26.1 Composition. Composition II shall be used only when local air pollution regulations governing the application of varnish precludes the use of composition I, When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with South Coast Air Quality Management District, Rule 442.

3.27 Workmanship.

3.27.1 Fabrication. Metal used in the fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall

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not reduce the strength of the steel to a value less than intended by the design. All bends shall be made by controlled means to ensure uniformity of size and shape. Surfaces shall be free of burrs, sharp edges or sharp corners (see 3.9).

3.27.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Weldments, including weld beads, shall be of homogeneous appearance and form without spattering or other irregularities, complying with the size, configuration and other dimensional requirements to develop the full strength of the parts joined by the welds. No cracks of any type are permitted. All welding is to be done by either AWS or ASME code certified welders. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings. Welds shall be in accordance with AWS D1.1.

3.27.3 Welders and welding operators. Welder and welding operators assigned to manual welding work covered by this specification shall be qualified in accordance with AWS D1.1 or the ASME Boiler and Pressure Vessel code for the materials joined and the type of welding operation to be performed. The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes (see 6.8).

3.27.4 Machine work. Tolerance and gauges for metal fit shall conform to commercial industry standards.

3.28 Government-furnished property. When specified (see 6.2), the following property in the quantity indicated will be furnished by the Government (see 6.6):

Item No.	Description	Quantity
1	Camouflage paint pattern	As required

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order and unless disapproved by the Government, the contractor's own facilities or any other, suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall

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inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.3 Acceptability criteria. Pumping assemblies which conform to all requirements in sections 3 and 5 and pass all applicable examinations and tests in section 4 of this document will be considered acceptable by the Government.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3 and 4.7.1).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.7).

4.3 First article inspection. First article inspection shall be performed on one complete unit when a first article sample is required (see 3.2 and 6.2). This inspection shall include the examination of 4.5 and tests of 4.6.1 and 4.6.2.

4.4 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.5 and the tests of 4.6.1. Quality conformance inspection shall be conducted on each pumping assembly.

4.5 Examination. Each pumping unit shall be examined as specified herein for the following defects. Presence of one or more defects shall be cause for rejection.

101. Parts or components missing or not functioning (see 3.1).
102. Assembly incorrect or not complete (see 3.1).
103. Materials not as specified (see 3.3).
104. Materials conforming to 21CFR177 and compounds listed in section 177.1020, 177.1030, 177.1040, 177.1050, 177.1480 of 21CFR not used as specified (see 3.3).
105. Materials are not corrosion resistant or treated to be corrosion resistant for the applicable storage and operating environment (see 3.3.1).
106. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).

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107. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the pumps (see 3.3.3) .
108. Use of cadmium plating not as specified (see 3.3.4).
109. Pump, engine and components are not contractor's latest models (see 3.4).
110. Threads not conforming to FED-STD-H28 (see 3.6).
111. Human factors engineering not as specified (see 3.9).
112. Pump case and volute not as specified (see 3.11.2.1 and 3.11.2.2).
113. Impeller not as specified (see 3.11.2.3).
114. Seals not as specified (see 3.11.2.4).
115. Priming port not as specified (see 3.11.2.5).
116. Piping and fittings not as specified (see 3.11.2.6).
117. Pump accessories not as specified. Each incorrect accessory shall constitute one defect (see 3.11.3 and 3.11.4).
118. Engine not as specified (see 3.12).
119. Automatic pump controller not as specified (see 3.13).
120. operator controls not as specified (see 3.13.1).
121. Hardware not as specified (see 3.13.2).
122. Pressure lines not as specified (see 3.13.3).
123. Instrument panel not as specified (see 3.14).
124. Wiring not as specified (see 3.14.1).
125. Fuel system not as specified (see 3.15).
126. Wheel mounting not as specified (see 3.16).
127. Tires and wheels not as specified (see 3.16.1).
128. Brakes not as specified (see 3.16.2).
129. Towing extension not as specified (see 3.16.3).
130. Stabilizing legs not as specified (see 3.16.4).
131. Lights not as specified (see 3.16.5).
132. Electrical connections not as specified (see 3.16.5.1).
133. Engine/pump housing not as specified (see 3.17).
134. Acoustic absorptive material and perforated metal cover not as specified (see 3.17.1).
135. Rock guard not as specified (see 3.18).
136. Transportability not as specified (see 3.19 thru 3.19.4).
137. Grounding assembly not as specified (see 3.20).
138. Identification not as specified (see 3.21.1 thru 3.21.2.1).
139. Instruction plates not as specified (see 3.21.3).
140. Performance curve plate not as specified (see 3.21.4).
141. Shipping data plate not as specified (see 3.21.5).
142. Lubrication not as specified (see 3.22).
143. Lubrication fittings not as specified (see 3.22.1).
144. Pressure-release device not as specified (see 3.22.2).
145. Treatment and painting not as specified (see 3.23).
146. Color not as specified (see 3.23).
147. Camouflage pattern data not as specified (see 3.23.1).
148. Battery box not as specified (see 3.24).
149. Battery not as specified (see 3.24).
150. Fungus and moisture treatment not as specified (see 3.26 and 3.26.1).
151. Workmanship not as specified (see 3.27 thru 3.27.4}.
152. Welders or welding operators not qualified as specified (see 3.27.3).

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4.6 Tests. When a first article test is required, the first article shall be subjected to the tests specified in 4.6.1 and 4.6.2. Failure to pass any phase of the test shall be cause for rejection.

4.6.1 Hydrostatic. The pump and fittings shall be subjected to hydrostatic test pressure of 1.5 times the shut off pressure at rated speed, for a period of 15 minutes, with both suction and discharge valves closed for a period of not less than 5 minutes. Any leakage, cracking, or permanent deformation of the assembly shall constitute failure of the test.

4.6.2 Performance.

4.6.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed in accordance with the applicable provisions of the test code for centrifugal pumps in the Hydraulic institute Standards for centrifugal, rotary and reciprocating pumps. All performance data shall be corrected to standard sea level conditions of a barometric pressure of 29.92 inches of mercury and water at 60 °F having a specific gravity of 1.0. Water at a temperature of between 60 and 80 °F may be used as the test fluid.

4.6.2.2 Pumping. The pump shall be operated as specified herein to determine the performance characteristics. The resultant data shall be used to plot the performance curve plate.

4.6,2.3 Priming test. The pumping assembly shall be tested on 10-foot static suction lift and shall, after initially filling the pump case, fully prime and deliver rated capacity within 2 minutes or less using a 6 inch diameter suction hose. Failure to prime and deliver rated flow in 2 minutes shall constitute failure of the test.

4.6.2.4 Operational test and functional test. The pumping assembly shall be operated for 4 hours at the rated conditions specified in 3.11.1. The pumping assembly shall be operated as required during this period to verify the functional operation of the controls, and the pump shall be examined during the operation for leakage through the pump casing or the shaft seals and for malfunction of any component. Any leakage attributable to defects in design, workmanship, materials, or to the malfunction of any component, or inability of the pump to deliver the minimum capacities specified herein shall constitute failure of this test.

4.6.2.5 Tilted position. The pumping assembly shall be operated for not less than 30 minutes while it is positioned 15 degrees from level on either side, the front and rear directions. Evidence of malfunction or misalignment of components shall constitute failure of this test.

4.6.2.6 Safety. The pumping assembly shall be evaluated for safety requirements throughout testing as specified in 4.6.2. Nonconformance to 3.8 shall constitute failure of this test.

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4.6.2.7 Human factors engineering inspection. The human factors engineering characteristics requirements shall be inspected during first article testing. Nonconformance to 3.9 shall constitute failure of this test.

4.6.2.8 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 11. As a minimum, noise levels shall be measured when equipment is operating under full load. Noise levels shall be measured at not fewer than 12 equal (horizontal) arc increments for the occasionally occupied positions. Additionally. the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with 3.10 provisions shall constitute failure of this test.

4.6.2.9 Electromagnetic interference. The first article shall be tested to determine conformance with 3.25. Nonconformance to 3.25 shall constitute failure of this test.

4.6.2.10 Tiedown provisions test. The tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.19.1. Inability to meet the requirements of 3.19.1 shall constitute failure of this test.

4.6.2.11 Slinging provisions test. The slining provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.19.2. Inability to meet the requirements of 3.19.2 shall constitute failure of this test.

4.6.2.12 Air transportability and airdrop test. The air transportability and airdrop shall be tested in accordance with MIL-STD-1791 to prove conformance to 3.19.3. Inability to meet requirements of 3.19.3 shall constitute failure of the tests.

4.6.2.13 Rail transportability test. The first article pumping unit shall be tested in accordance with MIL-STD-810 to prove conformance to 3.19.4. Inability to meet the requirements of 3.19.4 shall constitute failure of this test.

4.6.2.14 Ease of maintenance. The pumping assembly shall be evaluated for maintenance accessibility throughout testing. Nonconformance to 3.5.1 shall constitute failure of this test.

4.6.2.15 Reliability. Using the MTBF specified in 3.5.2, the pumping unit shall be tested for reliability. With the pump operating at the condition of 600 gpm and 350 feet total head, test until an “accept” or “reject” decision is reached, in accordance with figure 6. The θ_1, will be 100 hours and θ_0 will be 200 hours. A failure is defined as any malfunction which cannot be corrected within 30 minutes (all waiting periods are excluded) by adjustment, repair or replacement action, using controls and on-equipment tools or parts available to the using element, and is a malfunction which causes or may cause one or more of the following:

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a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
b. Damage to the pumping unit by continued operation.
c. Safety hazards to personnel,

Entering the reject region on figure 1 shall constitute failure of this test.

4.6.2.16 Environmental.

4.6.2.16.1 High temperature. The pumping assembly shall be tested as specified in MIL-STD-810, method 501, procedures I and II. The maximum ambient storage temperature shall be +160 °F, and the maximum operating temperature shall be +135 °F. The storage period of procedure I shall be 24 hours. After temperature stabilization at 135 °F and using constant temperature exposure, the operating period of procedure II shall be 3 hours. The pump assembly shall be operated at rated capacity. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.16.2 Low temperature. The pumping assembly shall be tested as specified in MIL-STD-810, method 502, procedures I and II. The minimum ambient storage temperature shall be -65 °F, and the minimum ambient operating temperature shall be -25 °F for the class II pumping assembly and +32 °F for the class I pumping assembly. The storage period of procedure I shall be 24 hours. After temperature stabilization at -25 °F for the class II pumping assembly or +32 °F for the class I pumping assembly and using constant temperature exposure, the operating period of procedure II shall be 3 hours. The pumping assembly shall be operated at rated capacity. Nonconformance to 3.7 shall constitute failure of this test.

4.7 Inspection of packaging.

4.7.1 First article inspection of packaging. The first article packaging shall be examined and tested in accordance with the first article pack inspection provisions of MIL-P-10603, and for the additional examination defects specified in 4.7.2.1.

4.7.2. Quality conformance inspection. Packaging of pump assemblies shall be examined in accordance with the quality conformance inspection provisions of MIL-P-10603, and for the additional examination defects specified in 4.7.2.1.

4.7.2.1 Examination. Sample packs, selected in accordance with the sampling requirements of MIL-P-10603, shall be examined for the following defects, in addition to the defects listed in MIL-P-10603. One or more defects shall be cause for rejection.

153. Pump preservative not as specified (see 5.2.1 and 5.2.2).
154. Level of packaging not as specified (see 5.2.1 and 5.2.2).
155. Depreservation guides not preserved as specified (see 5.3).
156. Marking not as specified (see 5.2.1 and 5.2.2).

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5. PACKAGING

5.1 First article pack. Unless otherwise specified (see 6.2), the contractor shall furnish a first article pack for examination and testing within the time frame specified (see 6.2), in accordance with the preproduction pack requirements of MIL-P-10603.

5.2 Preservation, packing, and marking. Preservation, packing and marking shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Each pumping unit shall be preserved, packed and marked in accordance with the level A requirements of MIL-P-10603 except type P-14 shall be used in lieu of P-10, the cooling system of a liquid cooled engine shall be preserved by the water and antifreeze procedure specified in MIL-E-10062 using antifreeze conforming to MIL-A-46153 and the consolidation containers, when required, shall be in accordance with PPP-B-601, domestic type, style A with unnailed closure.

5.2.2 Commercial. Each pumping unit shall be preserved, packed and marked in accordance with ASTM D 3951 except the cooling system shall be preserved as specified in 5.2.1.

5.3 Depreservation guide. When depreservation guides are specified for type I pumping assemblies (see 6.2), one guide shall be placed in a waterproof envelope marked "Depreservation Guide" and shall be secured to the pumping assembly in a conspicuous location. The second guide shall be preserved with the technical publications and the bag shall be marked to indicate that the depreservation guide is inside.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The class I pumping assembly is intended to pump drinking water through a 6 inch hoseline in cold and hot temperatures, and is a component of the Tactical Water Distribution System. The class II pumping assembly is a flood and transfer pump intended to pump petroleum products to and from large temporary storage facilities in cold and hot temperatures, and is a component of the Inland Petroleum Distribution System. Operating conditions could be humid or dry with frequent periods of blowing sand.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class of pump required (see 1.2).
c. Issue of DoDISS to be cited in solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is not required for first article inspection and approval (see 3.2, 4.3 and 6.3).

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e. Color of finish coat if not as specified (see 3.23).
f. Time frame when camouflage pattern data is required (see 3.23.1).
g. When camouflage patterns will be provided (see 3.23.1 and 6.2.1).
h. When the property listed in 3.28 will be furnished by the Government (see 6.5).
i. Time frame required for submission of the first article pack (see 5.1).
j. When a first article pack is not required (see 5.1).
k. Degree of packaging required (see 5.2).
l. When depreservation guides are required (see 5.3 and 6.7).
m. When a transportability report is required (see 6.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

	Reference paragraph	DID Number	DID Title
a.	3.24.1	DI-MISC-80192	Color photograph prints
b.	3.24.1	DI-MISC-80176	Camouflage line art data
c.	3.19	DI-PACK-80880	Transportability report

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) , must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.5 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor's obligation to preserve, pack, and mark the pumping assembly in accordance with this specification.

6.6 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.28.

6.7 Depreservation guide. When necessary (see 5.3), the contracting officer should reqire the contractor to prepare two depreservation guides in accordance with DA Fern 2258 and arrange to furnish the necessary copies of DA Form 2258, when requested by the contractor.

6.8 Welders and welding operators. The contracting officer should consider requalifying welders or welding operators in the event of evidence of poor welds (see 3.27.3).

6.9 Subject term (key word) listing.

Tactical Water Distribution System
Pump, hoseline, drinking water
Pump, water
Inland Petroleum Distribution System
Pump, flood and transfer
Pump, POL

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 4320-A369

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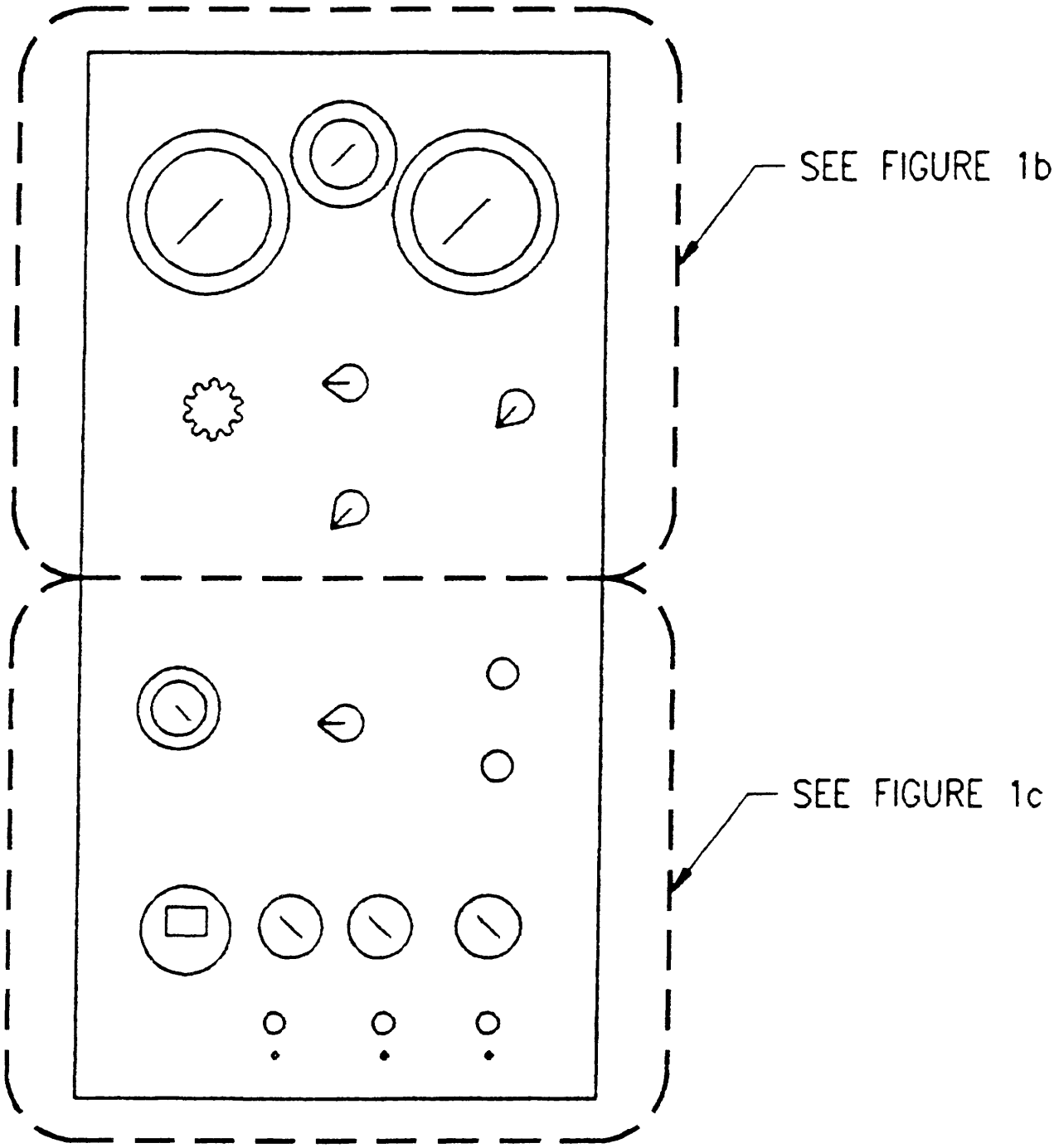
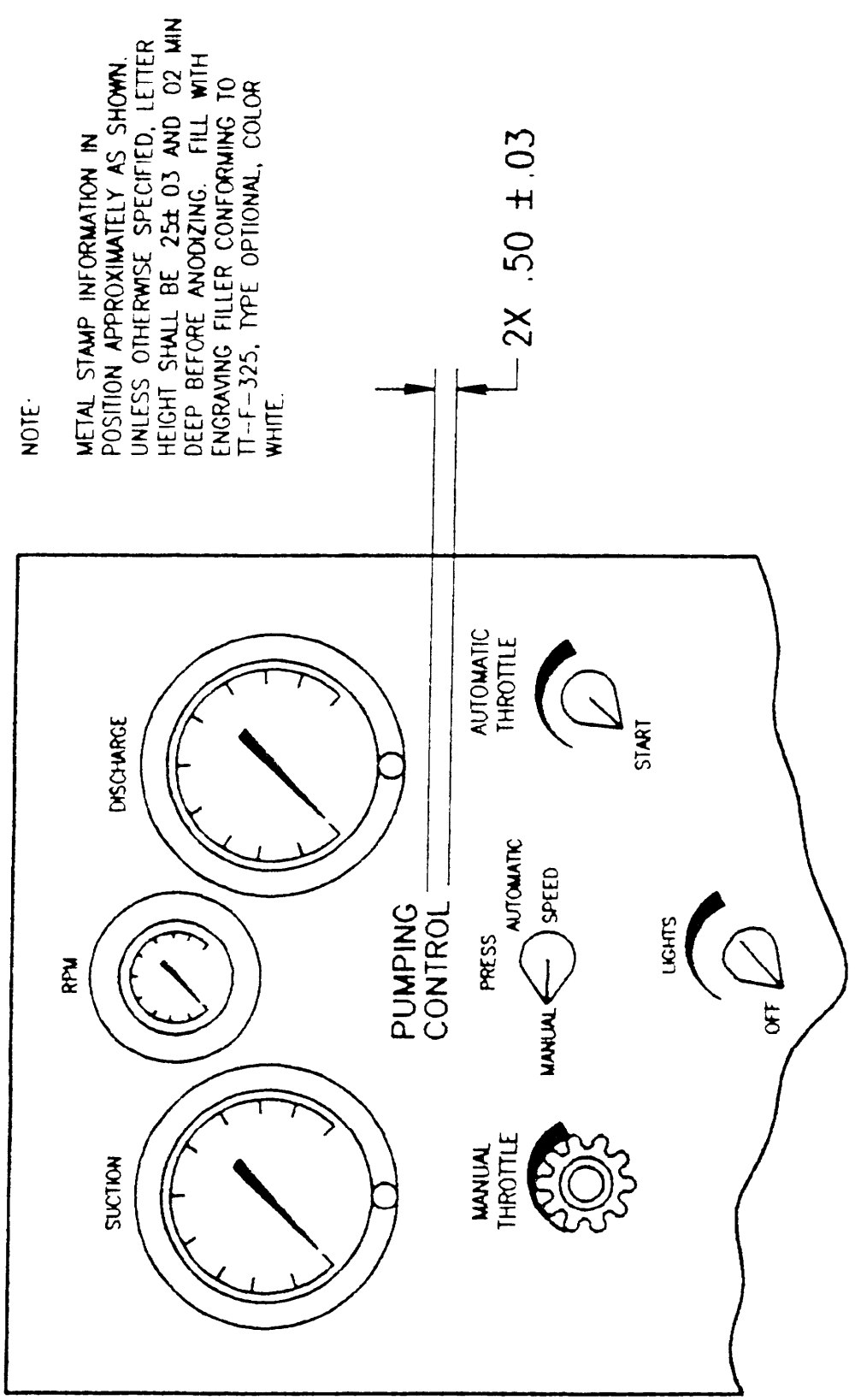


FIGURE 1a. Control and instrument panel.

X-5051

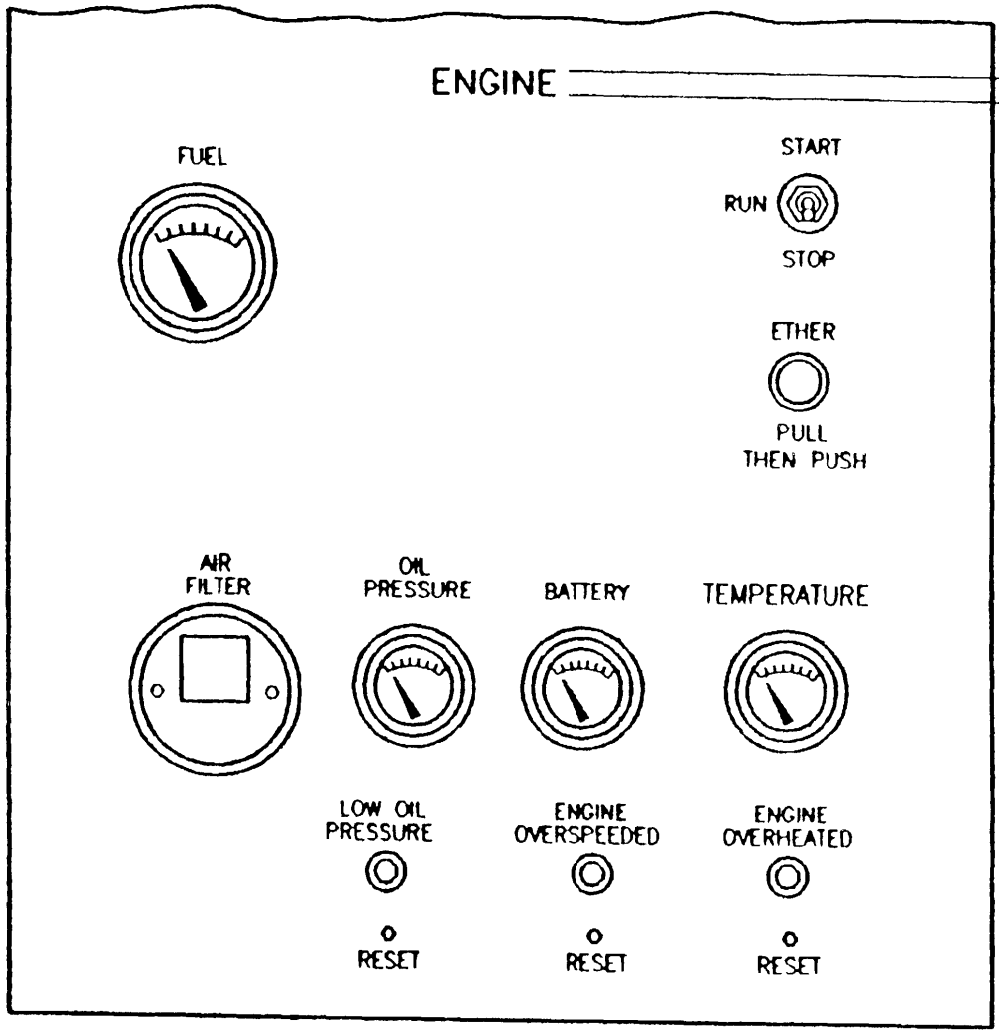
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X-5052

FIGURE 1b. Control and instrument panel.

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.50 ±.03

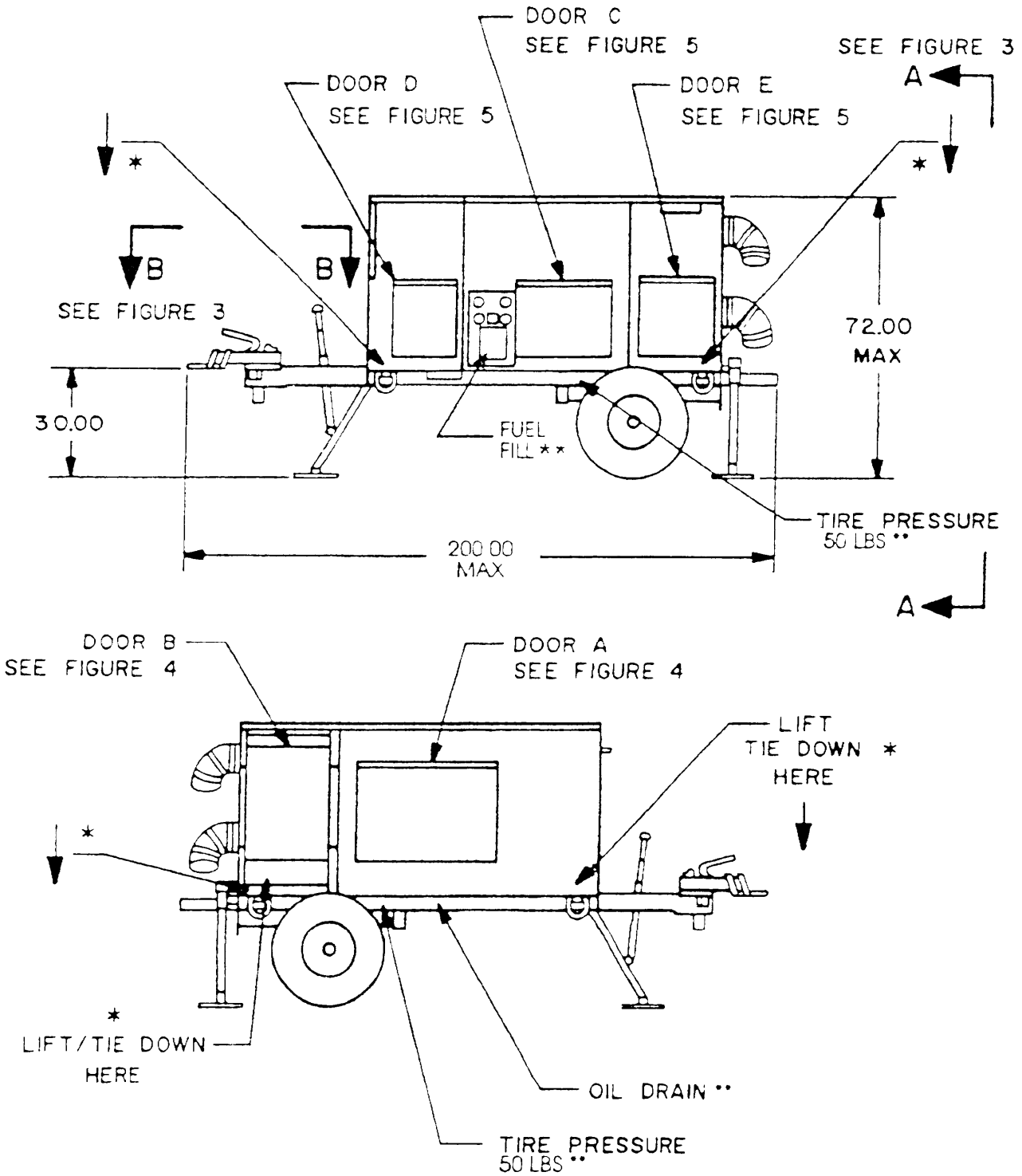
NOTE:

METAL STAMP INFORMATION IN POSITION APPROXIMATELY AS SHOWN. UNLESS OTHERWISE SPECIFIED, LETTER HEIGHT SHALL BE .25±.03 AND .02 MIN DEEP BEFORE ANODIZING. FILL WITH ENGRAVING FILLER CONFORMING TO TT-F-325, TYPE OPTIONAL, COLOR WHITE.

MIL-P-53051A(ME)

FIGURE 1c. Control and instrument panel. X-5053

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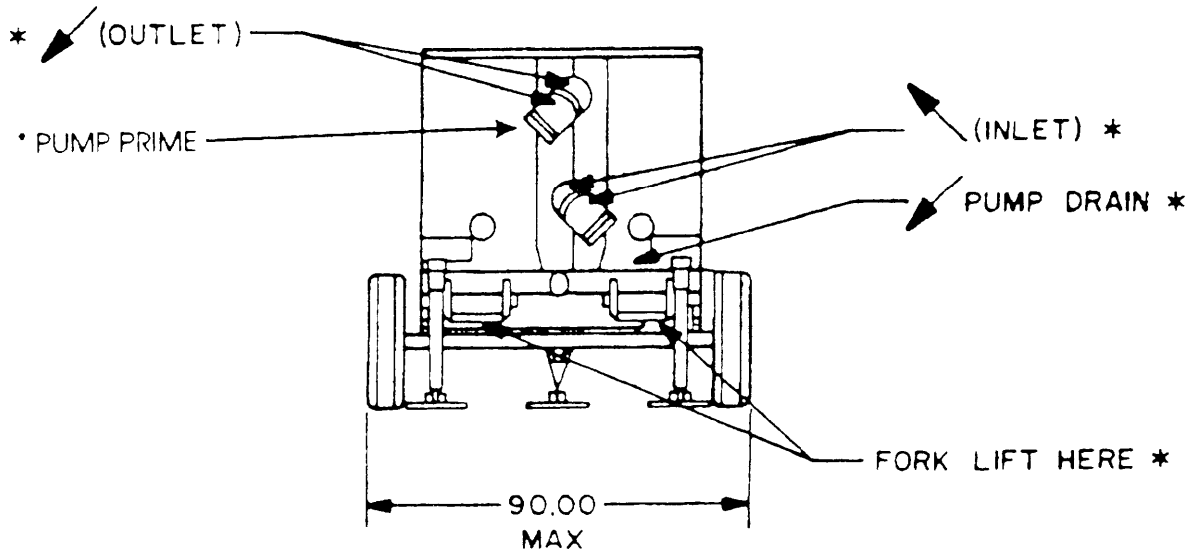


* STENCIL, IN 1 - INCH HIGH LETTERS ARROW SYMBOL AND DATA AT APPROXIMATE POSITION SHOWN. SEE FIGURE 4 FOR STENCIL SPECIFICATION

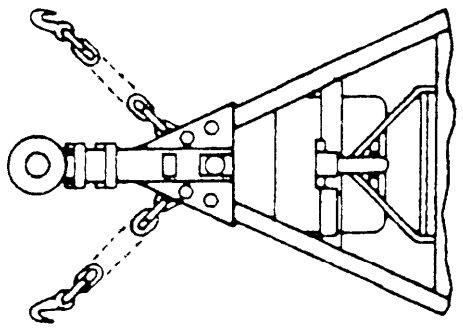
** STENCIL, IN 1/2 - INCH HIGH LETTERS DATA AT APPROXIMATE POSITION SHOWN SEE FIGURE 4 FOR STENCIL SPECIFICATION

FIGURE 2. 600 GPM pump. X-4777A

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VIEW A-A



VIEW B-B

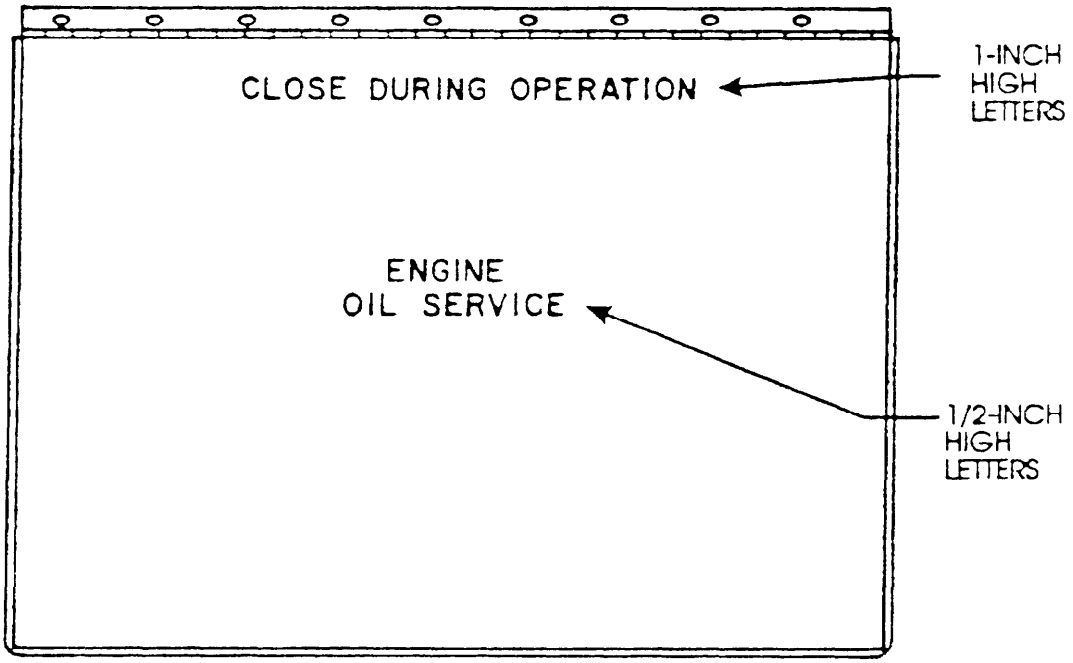
* STENCIL, IN 1/2 - INCH HIGH LETTERS, ARROW SYMBOL AND DATA AT APPROXIMATE POSITION SHOWN. SEE FIGURE 4 FOR STENCIL SPECIFICATION

FIGURE 3. 600 GPM pump.

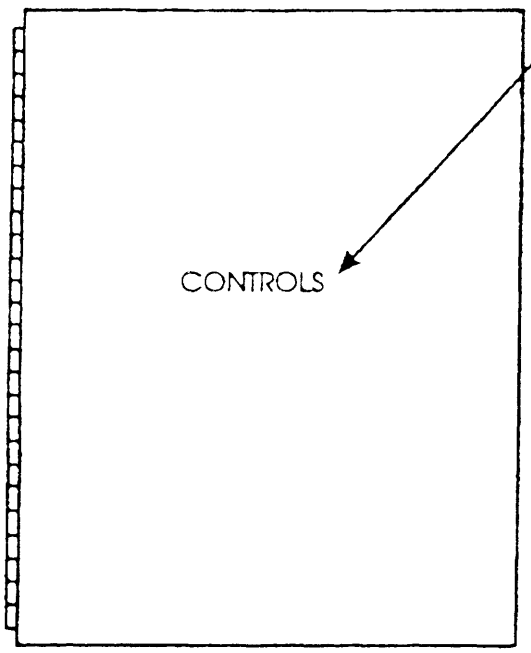
X-4778A

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DOOR A



DOOR B



STENCIL SPECIFICATIONS FOR
1 - INCH HIGH LETTERS

(FOR 1/2 - INCH HIGH LETTERS
THESE VALUES SHALL BE HALVED)

	(IN INCHES)
LETTER HEIGHT	1.000
LETTER WIDTH	.600
LETTERS "M" and "W" WIDTH	.800
LETTER "I" WIDTH	.160
STROKE WIDTH	.160
LETTER SPACING	.200
WORD SPACING	.800
LINE SPACING	.500

FIGURE 4. Door panel stenciling.

X-4779B

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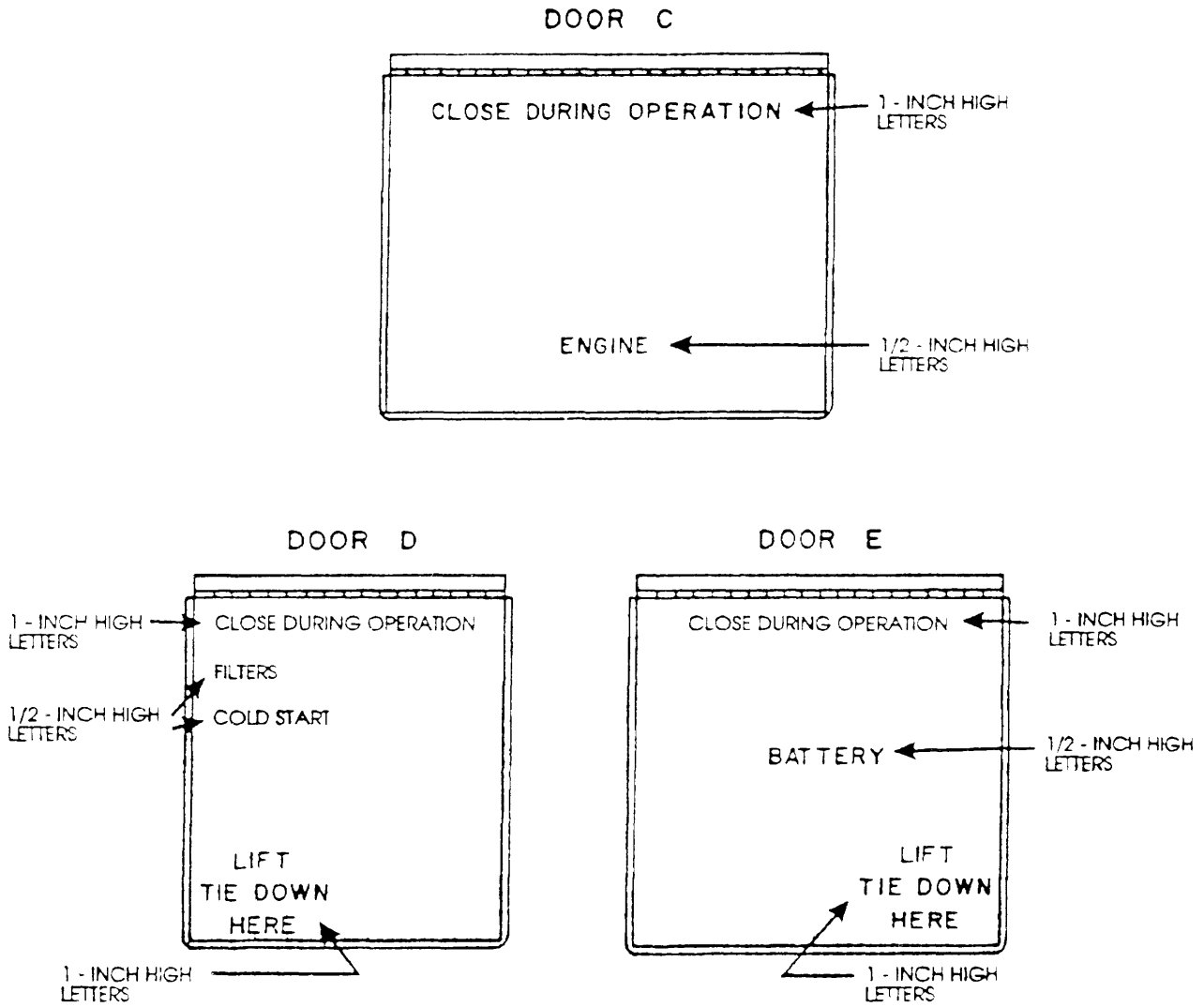


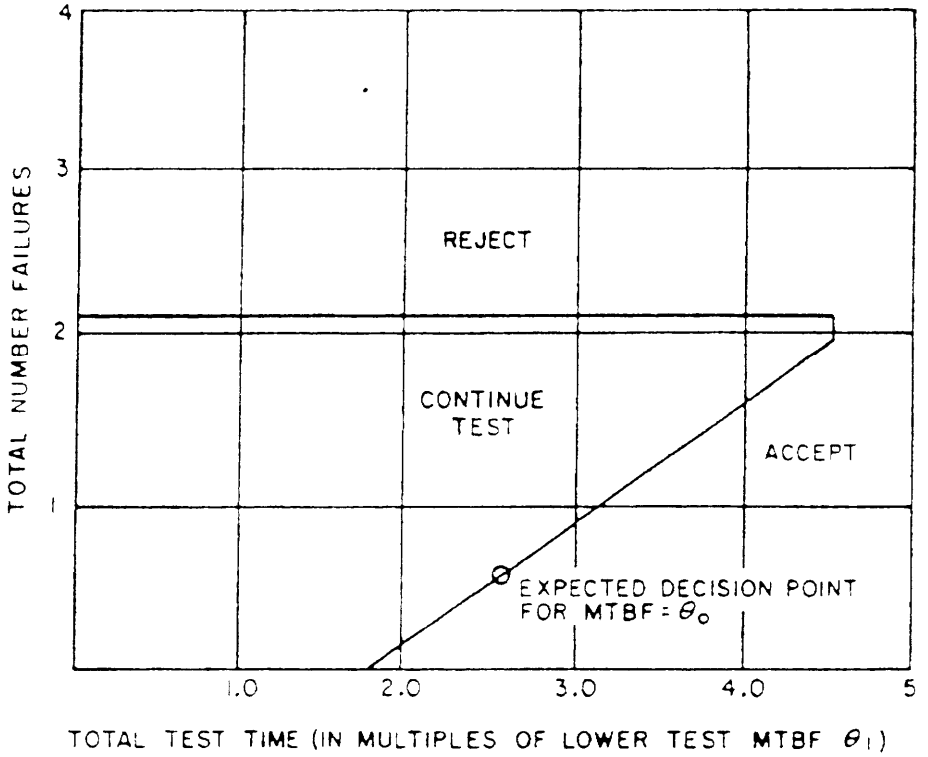
FIGURE 5. Door panel stenciling

X-4780A

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DECISION RISKS (NOMINAL) 30 PERCENT

DISCRIMINATION RATIO 2.0, 1



TOTAL TEST TIME *

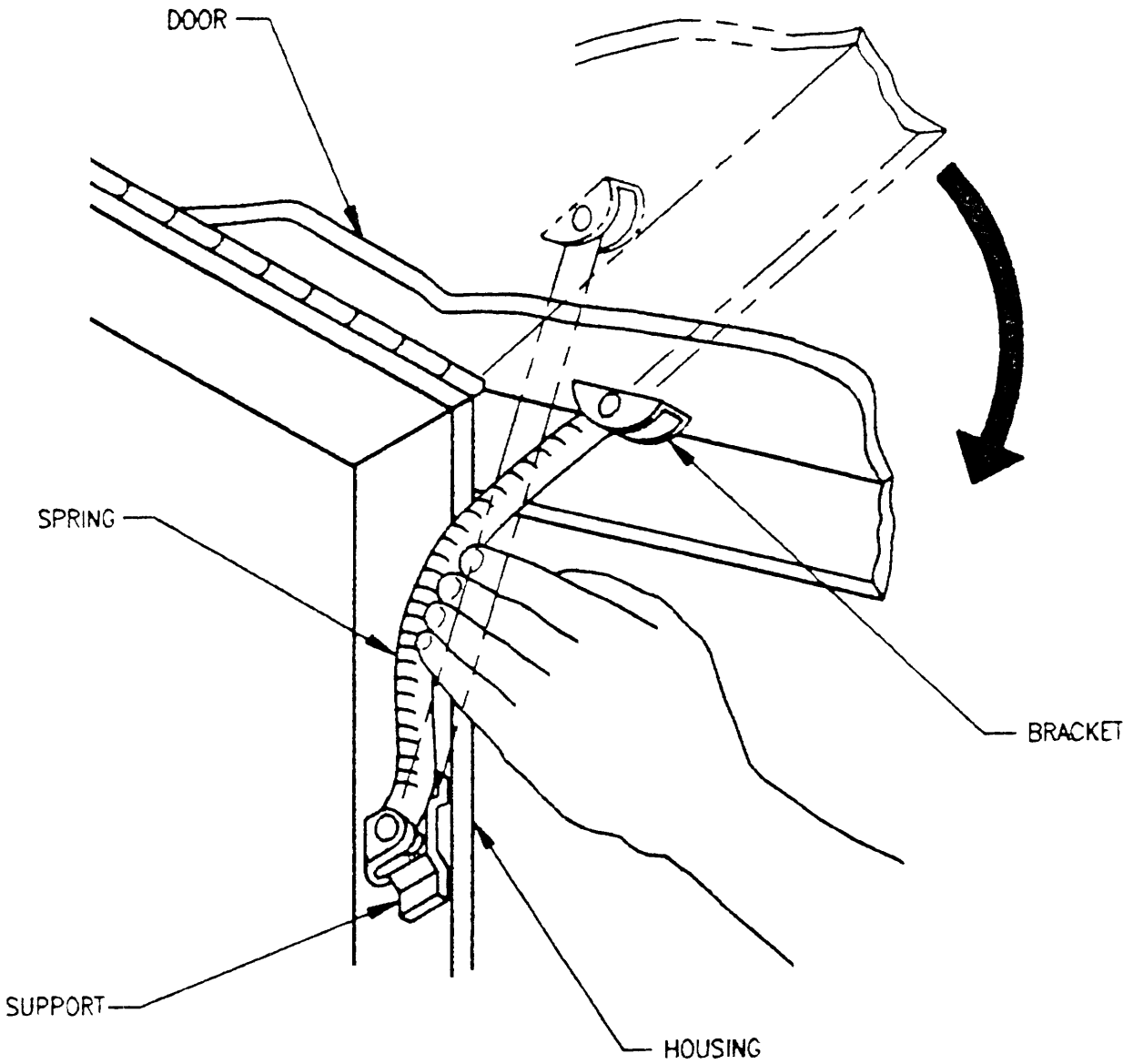
NUMBER OF FAILURES	REJECT (EQUAL OR LESS)	ACCEPT (EQUAL OR MORE)
0	N/A	1.72
1	N/A	3.10
2	N/A	4.50
3	4.5	N/A

* TOTAL TEST TIME IS TOTAL UNIT HOURS OF EQUIPMENT ON TIME AND IS EXPRESSED IN MULTIPLES OF THE LOWER TEST MTBF. REFER TO 4.6.2.12 FOR MINIMUM TEST TIME PER EQUIPMENT.

FIGURE 6. Accept-reject criteria.

X-4281A

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SUGGESTED SOURCES OF SUPPLY:

MOONLITE MARINE CORP
776 W 17TH ST
COSTA MESA CA 92627

EMERGENCY ONE INC
1701 SW 37TH AVE
OCALA FL 32678

FIGURE 7. Automatic door holder.

X-5054

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter be given.
   The submitter of this form must complete blocks 4, 5, 6, and 7.
   . The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-53051A(ME)	2. DOCUMENT DATE (YYMMDD) 940923

3. DOCUMENT TITLE PUMPING ASSEMBLY, WHEEL-MOUNTED, DIESEL-ENGINE-DRIVEN, SELF-PRIMING, CENTRIFUGAL, 600 GPM AT 350 FEET TOTAL HEAD AND 1250 GPM AT 180 FEET TOTAL HEAD

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (If applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468	(2) DSN 654-3468
c. ADDRESS (Include Zip Code) 'S ARMY BELVOIR RDE CTR ATTN SATBE TSE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340	DSN 289-234

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